Module-I
Introduction to Mechanical Engineering

- Introduction: Streams in mechanical engineering and their relevance/significance,
- Role of mechanical engineers in industry.
- Energy conversion: Introduction of water turbine and basic working principles of Pelton Turbine.
- Flying machines: Classification, basic parts involved in drone making, working principle of Drones.
- Refrigeration (VCR only) and Room air conditioning principles.

Introduction:
MECHANICAL ENGINEERING

- Mechanical Engineering is defined as the branch of engineering that deals with the design, development, construction, and operation of mechanical systems and tools. It include machines, tools, and equipment used in various industries, such as transportation, manufacturing, power generation, and medical devices etc.
- They work on power-producing machines such as electricity-producing generators, internal combustion engines, steam and gas turbines, and jet and rocket engines.
- They also develop power using machines such as refrigeration and air-conditioning equipment, robots used in manufacturing, machine tools, materials handling systems, and industrial production equipment.

Streams in mechanical engineering and their relevance/significance

1) **Design Engineering** → Product & machine design (CAD/CAE); ensures safety, reliability, and innovation.
2) **Thermal Engineering** → Heat transfer, power plants, HVAC(**Heating, Ventilation, and Air Conditioning**); vital for energy generation & conservation.
3) **Manufacturing/Production Engineering** → Machining, automation, quality control; backbone of industrial production.
4) **Automobile Engineering** → Vehicle design, EVs, alternative fuels; drives transportation innovation.
5) **Mechatronics & Robotics** → Integration of mechanics, electronics & AI; essential for automation & Industry 4.0.
6) **Industrial Engineering** → Productivity, lean systems, supply chain; optimizes efficiency and reduces waste.
7) **Aerospace Engineering** → Aircraft/spacecraft design, propulsion; key for aviation, defense, and space exploration.
8) **Materials & Metallurgy** → Advanced materials, composites, nanotech; improves strength, durability & performance.
9) **Marine Engineering** → Ship design, marine propulsion; crucial for naval defense & shipping industry.
10) **Renewable Energy & Sustainability** → Solar, wind, biomass; addresses climate change & energy crisis

Role of Mechanical Engineering in Industries

- Mechanical engineering is a foundational branch of engineering that applies the principles of physics, mathematics, and material science to design, analysis, manufacture, and maintain mechanical systems.
- It plays a crucial role in advancing industries and improving societal welfare.

1) **Power Generation**: Mechanical engineers design and develop power-generating machines such as internal combustion engines, gas turbines, and steam and wind turbines etc
2) **Heating and Cooling Systems:** They design and develop heating, ventilation, refrigeration and air conditioning systems for buildings and other structures.
3) **Transportation:** Mechanical engineers are involved in designing and developing transportation systems, including cars, trains, airplanes, steamers and boats.
4) **Industrial Equipment:** They design, develop and maintain industrial equipment such as machine tools, robots, and conveyor systems & belts
5) **Infrastructure:** Mechanical engineers play a key role in the design and maintenance of infrastructure, including buildings, bridges, roads, and transportation systems.

Energy conversion: Introduction of water turbine

- A water turbine is a rotary machine that convert hydraulic energy or hydro-potential into mechanical energy.
- Mechanical energy developed by turbines is used to run electric generators coupled to the shaft of turbines.
- Water turbines are mostly found in dams to generate electric power from water potential energy.
- Pelton wheel is impulse type water turbine which extracts energy from impulse of moving water when the water strikes the pelton cup at very high speed, it induces an impulsive force which makes the turbine rotate. In short the pelton wheel transforms the kinetic energy of water jet into rotational energy.

Basic working principles of Pelton Turbine.

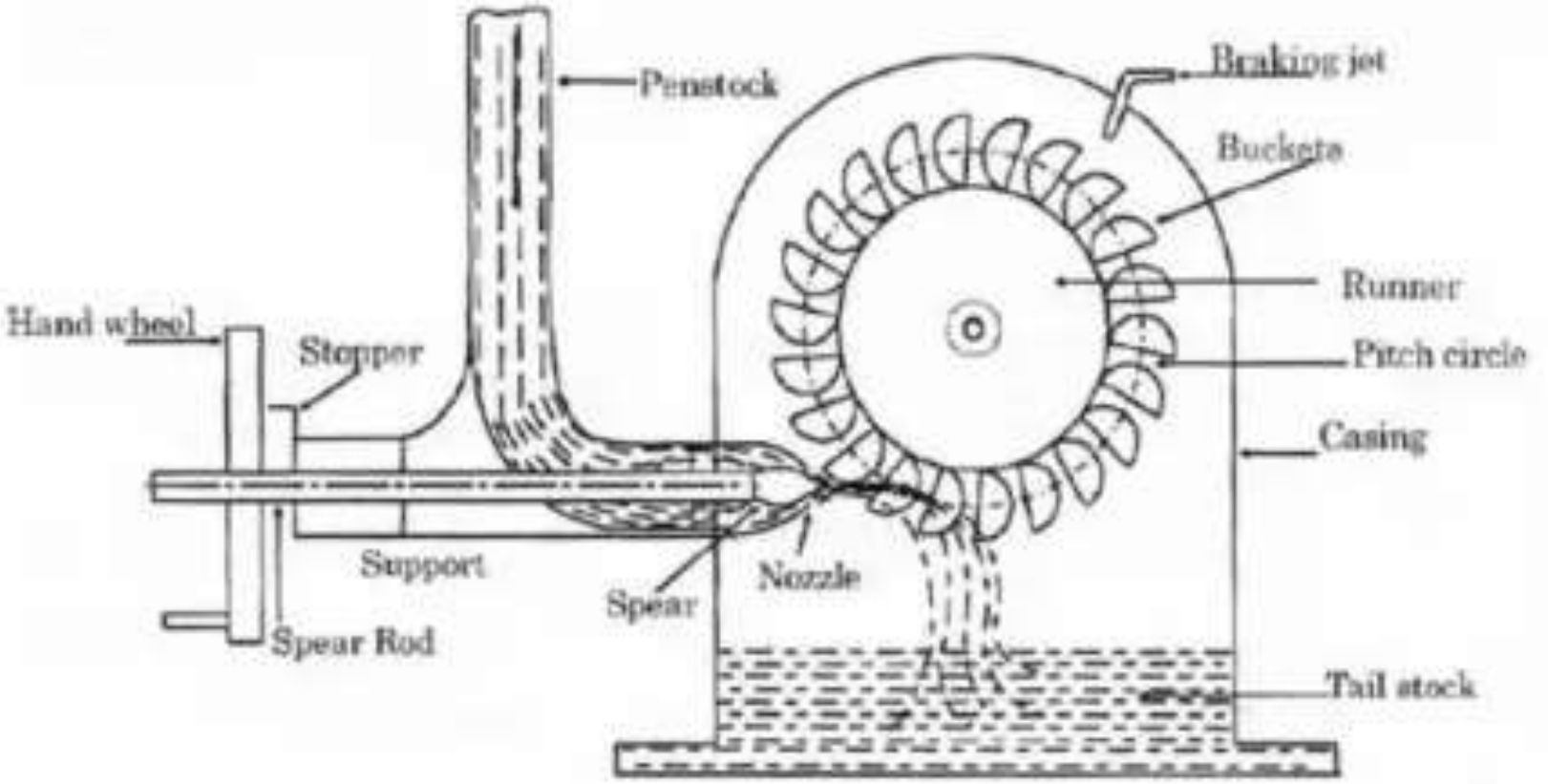


Figure 1- Schematic Diagram of Pelton wheel

The main parts of Pelton Turbine are:

1) **Nozzle and Flow regulation arrangement**: The amount of water flowing out from the nozzle is regulated by providing a spear in the nozzle. The spear is a conical shaped needle which is operated by manually or automatically in an axial direction depending upon the size of the unit.
2) **Runner and Buckets**: Runner consists of a circular disc on the periphery of which number of buckets are attached with equal spacing. The pelton buckets are in cup or bowl shape and each bucket is divided into symmetrical parts by a wall called splitter.
3) **Casing**: The function of casing is to prevent the splashing of water and to guide the discharged water to tail race.
4) **Breaking Jet**: When the nozzle is closed completely by moving the spear in forward direction, the amount of water striking the bucket reduces to zero. But due inertia, the runner goes on revolving for some more time. To stop the runner in a short time, a small nozzle is provided which directs the water from the back of the buckets.

Basic working principles of Pelton Turbine

- The water stored at a high head is made to flow through the penstock and reaches the nozzle of the Pelton turbine.
- The nozzle converts this pressure energy into a **high-velocity jet**.
- The water jet strikes the **double-cupped buckets** fixed on the runner wheel.
- The jet is **deflected**, and due to change in momentum, a **force** is exerted on the buckets.
- This force causes the **runner to rotate**, producing mechanical energy.
- The shaft of the runner is connected to a **generator**, which converts mechanical energy into **electrical energy**.

Flying machines: Introduction of Drone

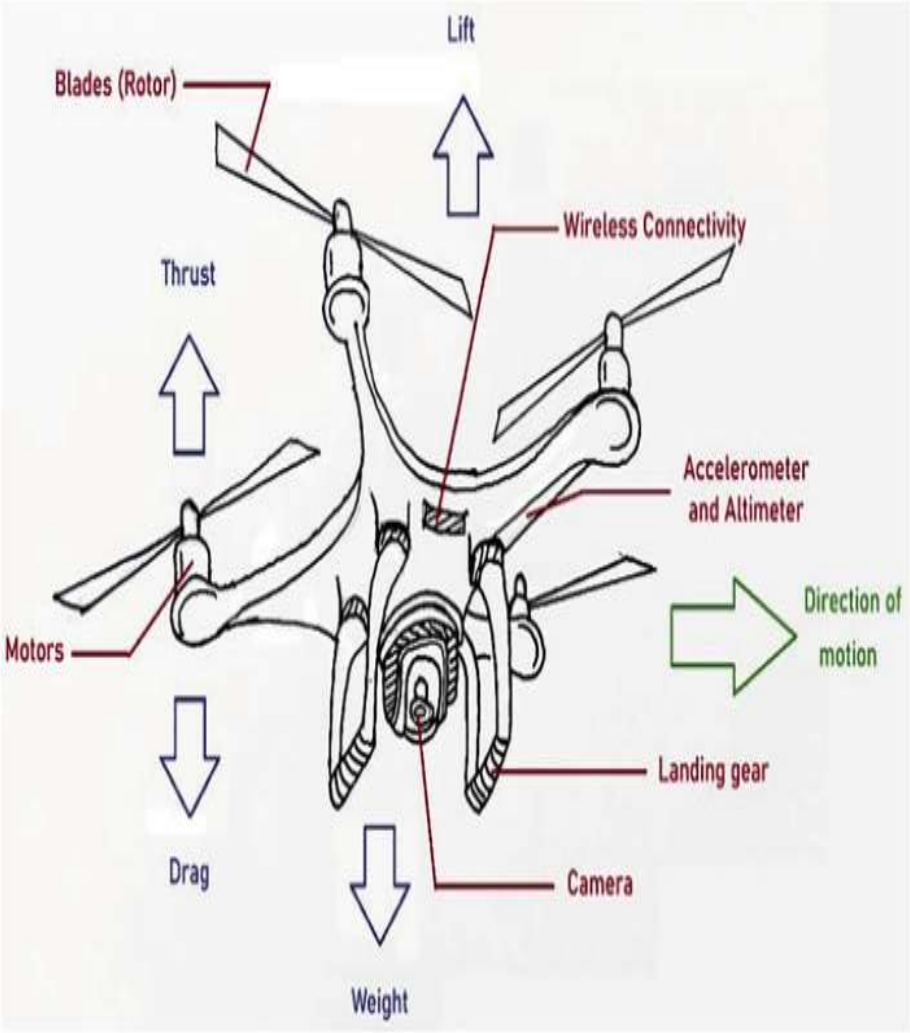
- A **drone** is an **unmanned aerial vehicle (UAV)** that can fly without a human pilot onboard, controlled either **remotely** by an operator or autonomously using onboard computers, sensors, and GPS.
- A drone is a **flying machine** that uses motors, propellers, and electronic systems to stay airborne, move in different directions, and perform tasks such as surveillance, photography, delivery, or military operations.

Flying machines: Classification of drone

- Drones can be classified based on **design, size, range, and applications**:
- **a) Based on Design:**
- **Multi-Rotor Drones** – Common type, Used for aerial photography, surveillance.
- **Fixed-Wing Drones** – Airplane-like design, long endurance, used for mapping and long-distance flights.
- **Single Rotor Drones** – Like helicopters, higher endurance and lifting capacity.
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- **b) Based on Range:**
- **Short Range (up to 5 km)** – Hobby drones.
- **Medium Range (5–50 km)** – Used for agriculture, monitoring.
- **Long Range (>50 km)** – Military, cargo, and surveillance drones.
- **c) Based on Use:**
- **Civilian Drones** – Photography, delivery, agriculture, inspection.
- **Military Drones** – Reconnaissance, combat, surveillance.
- **Commercial/Industrial Drones** – Mapping, mining, construction monitoring.

Basic Parts Involved in Drone Making

- A drone generally consists of the following **main components**:
- **Frame (Chassis)** – Structure that holds all components together (usually lightweight carbon fiber/plastic).
- **Motors** – Provide thrust; quadcopters typically use 4 brushless DC motors.
- **Propellers** – Convert motor rotation into lift and thrust.
- **Electronic Speed Controllers (ESCs)** – Control motor speed, direction, and braking.
- **Battery (Li-Po)** – Power source.
- **Flight Controller (FC)** – The "brain" of the drone; stabilizes and controls flight using sensors.
- **Sensors** – IMU (Inertial Measurement Unit) (gyroscope, accelerometer), GPS, barometer, magnetometer.
- **Transmitter & Receiver (Radio System)** – For communication between pilot and drone.
- **Camera** – For aerial photography or surveillance.
- **Landing Gear** – Supports safe landing.



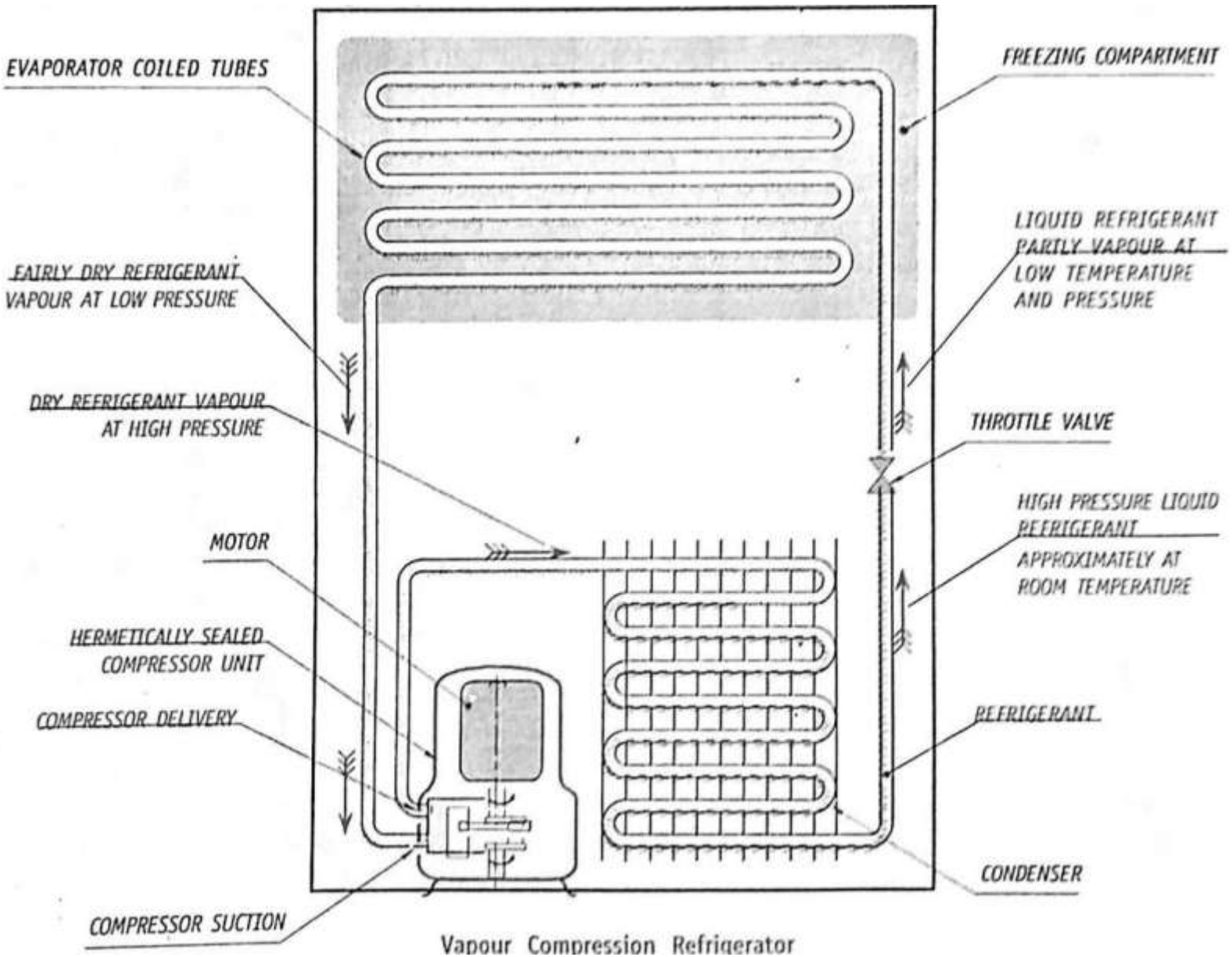
working principle of drone

- The working principle of drone operations is based on the balance of forces: **lift, weight, thrust, and drag**. Drones achieve flight by generating lift through their propellers or rotors, which counteracts gravity. Thrust is generated when the propellers spin at high speeds, pushing air downwards and lifting the drone upwards.
- Simultaneously, the motors adjust the rotational speed of each propeller to control roll and pitch, allowing the drone movement in three-dimensional space.
- The flight controller - often called the "brain" of the drone - interprets input from sensors and GPS modules. It then adjusts motor speeds to stabilize the drone and follow pilot commands or autonomous pre-planned missions.

INTRODUCTION

- Refrigeration is defined as a method of reducing the temperature of a system below that of the surrounding and maintains it at the lower temperature by continuously abstracting the heat from it.
- Principle of refrigeration :- The heat is to be removed continuously from a system at lower temperature and transfer it to the surroundings at higher temperature.
- Refrigerators work on the second law of thermodynamics. In the process of refrigeration, unwanted heat is taken from one place and discharged into another.

Working principle of **VAPOUR COMPRESSION REFRIGERATOR**
(VCR)refrigeration system



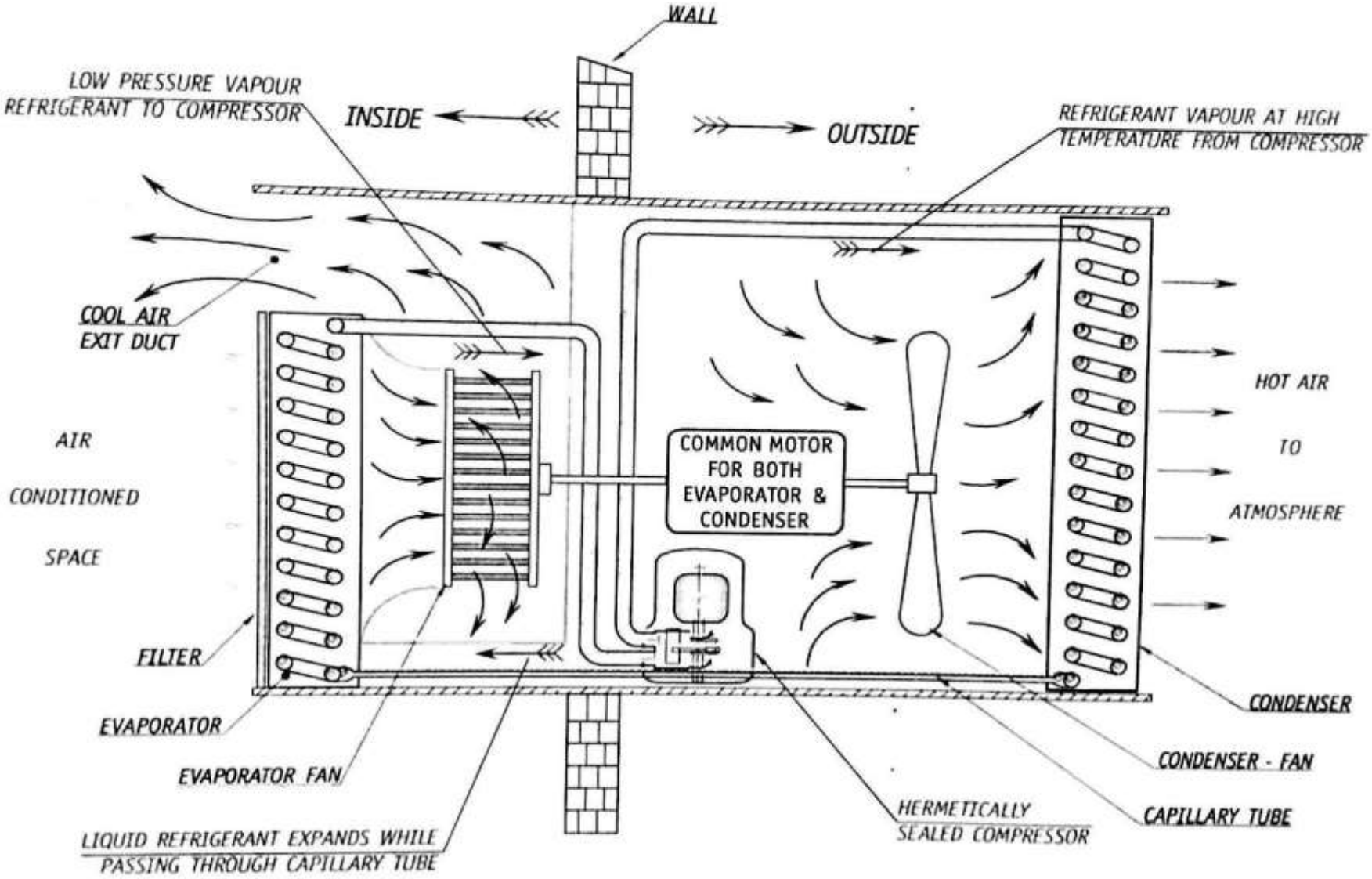
Working principle of VCR refrigeration system

- It mainly consists of a compressor, a throttle valve,a condenser and an evaporator made of coiled tubes installed in the freezing compartment of the refrigerator.
- The refrigerant at low pressure and temperature passing in the evaporator coiled tubes absorbs heat from the contents in the freezing compartment and evaporates.
- This lowers the temperature of freezing compartment.
- The vapour refrigerant at low pressure from evaporator is drawn by the compressor which compresses it to high pressure.
- This increase in pressure increases the saturation temperature of the refrigerant higher than the temperature of the cooling medium (atmospheric air) in the condenser so that vapour can reject heat in the condenser.
- In the condenser it gives off its latent heat to the atmosphere air and condenses to liquid.
- The high pressure liquid refrigerant now flows to the throttle valve in which it expands to a low pressure.
- Temperature reduces to -10°C and vapour will be wet.
- This wet vapour now passes to the evaporator coils where it absorbs heat from the surrounding and the cycle repeats.
- Thus heat is continuously removed from the contents of the refrigerator in the evaporator and rejected in the condenser to the atmospheric air.
- This will keep the contents of the refrigerator at lower temperature.
- The most commonly used refrigerant in vapour compression refrigerator is dichlorodifluoromethane, popularly known as Freon12.

AIR CONDITIONING

- Providing a cool pleasing indoor atmosphere by cooling, humidifying or dehumidifying, cleaning and recalculating the surrounding air is called air conditioning.
- The artificial cooling of air and conditioning it to provide maximum comfort to human beings is called comfort air conditioning.
- The artificial cooling of air and conditioning it to provide a controlled atmosphere required in some engineering, manufacturing and processing is called industrial air conditioning.

working principle of room air conditioner



Air Conditioner

working principle of room air conditioner

- It mainly consists of an evaporator, condenser, compressor, two fans one each for evaporator and condenser units usually driven by the single motor, capillary, etc. It is generally mounted on a window sill such that the evaporator unit is inside the room and the condenser part projecting outside the building.
- The high pressure, high temperature liquid refrigerant from the condenser is passed to the evaporator coils through the capillary tube where it under goes expansion.
- The refrigerant in evaporator coils absorbs heat from the air passing over it from the interior and evaporates.
- The high temperature evaporated refrigerant is compressed to high pressure by a compressor and delivered to the condenser, where it is cooled or condensed to liquid by giving off the heat to the atmospheric air passing over it.
- The cooled high pressure refrigerant now passes through the capillary tube where it under goes expansion and again re-circulated to repeat the cycle continuously.